

explanation

$$\mathbf{nordWAR} \text{ (Position Players)} = \frac{wRAA + wSB + \text{Fielding} + \text{PosAdj} - R_{\text{rep}}}{RPW}$$

$$\mathbf{nordWAR} \text{ (Pitchers)} = \text{pitchWAR}$$

$$\text{pitchWAR} = \left(\frac{RAAP9}{dRPW} + \text{RepLevel} \right) \cdot \frac{IP}{9}$$

$$wRAA = \frac{wOBA - wOBA_{\text{lg}}}{wOBA \text{ scale}} \cdot PA$$

$$wOBA = \frac{w_{BB}BB + w_{HBP}HBP + w_{1B}1B + w_{2B}2B + w_{3B}3B + w_{HR}HR}{PA + BB + SF + HBP}$$

$$wSB = SB \cdot r_{SB} + CS \cdot r_{CS} - \text{lgwSB} \cdot (1B + BB + HBP)$$

$$\text{Fielding} = (FPct - FPct_{\text{lg}}) \cdot 0.8 \cdot G$$

$$\text{PosAdj} = \frac{G}{G_{\text{team}}} \cdot \text{PosRunValue} \cdot \frac{G_{\text{season}}}{162}$$

$$R_{\text{rep}} = 10 \cdot \frac{G_{\text{season}}}{162} \cdot \frac{G_{\text{played}}}{G_{\text{season}}}$$

$$RPW = 2 \cdot \left(\frac{\text{Total Runs}}{\text{Teams} \cdot G_{\text{played}}} \right)$$

$$FIP = \frac{13HR + 3(BB + HBP) - 2K}{IP} + C_{\text{FIP}}$$

$$RAAP9 = FIPR9_{\text{lg}} - FIPR9$$

$$dRPW = \left(\frac{(18 - IP/G) \cdot FIPR9_{\text{lg}} + (IP/G) \cdot FIPR9}{18} + 2 \right) \cdot 1.5$$

$$\text{RepLevel} = 0.03 \left(1 - \frac{GS}{G} \right) + 0.12 \left(\frac{GS}{G} \right)$$

Definitions

1. $wRAA$: Batting runs above league average.
2. $wOBA$: Weighted on-base average.
3. wSB : Stolen base runs above average.
4. Fielding: Defensive value vs league-average fielding percentage.
5. PosAdj: Positional adjustment.
6. R_{rep} : Replacement-level runs.
7. RPW : Runs per win.
8. FIP : Fielding Independent Pitching.
9. $RAAP9$: Runs above average per 9 innings.
10. $dRPW$: Dynamic runs per win.
11. RepLevel: Starter/reliever adjustment.
12. IP : Innings pitched.